



JKBOSE 2026

CLASS 11TH

SETS

MATHS

LECTURE-07

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TOPICS TO BE COVERED

1

Exercise Discussion 1.4

2

Complement of a Set

3

Exercise Discussion 1.5

4

Related Problems

* Miscellaneous Problems

operation on sets:

$$* A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$* A \cap B = \{x : x \in A \text{ \& } x \in B\}$$

$$* A - B = \{x : x \in A \text{ but } x \notin B\}$$

EXERCISE – 1.4

Question



Find the union of each of the following pairs of sets:

(i) $X = \{1, 3, 5\}$ $Y = \{1, 2, 3\}$ (Sol) $X \cup Y = \{1, 2, 3, 5\}$

(ii) $A = \{a, e, i, o, u\}$ $B = \{a, b, c\}$ (Sol) $A \cup B = \{a, b, c, e, i, o, u\}$

(iii) $A = \{x : x \text{ is a natural number and multiple of } 3\}$ (iii) (Sol) $A = \{3, 6, 9, 12, 15, \dots\}$
 $B = \{x : x \text{ is a natural number less than } 6\}$
 $B = \{1, 2, 3, 4, 5\}$

H/W (iv) $A = \{x : x \text{ is a natural number and } 1 < x \leq 6\}$
 $B = \{x : x \text{ is a natural number and } 6 < x < 10\}$

(v) $A = \{1, 2, 3\}, B = \phi \Rightarrow A \cup \phi = A$

$\Rightarrow A \cup B = \{1, 2, 3, 4, 5, 6, 9, 12, 15, \dots\}$

Question

Let $A = \{a, b\}$, $B = \{a, b, c\}$. Is $A \subset B$? What is $A \cup B$?

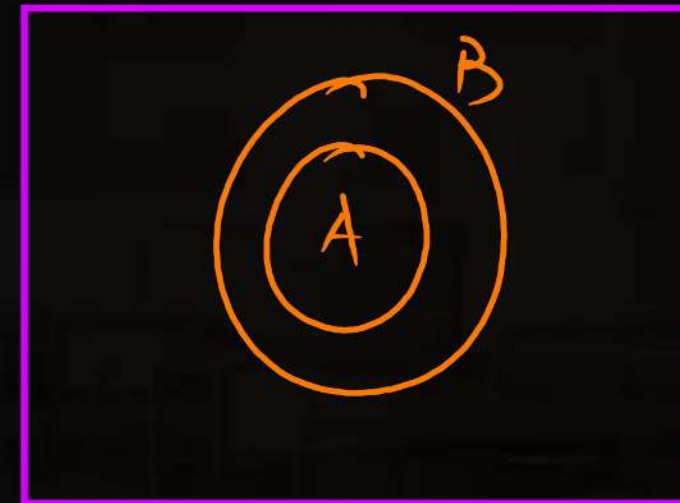
(Sol) Yes & $A \cup B = \{a, b, c\}$

$$\Rightarrow A \cup B = B.$$

Question

Let A and B are two sets such that $A \subset B$, then what is $A \cup B$?

(Sol) $A \cup B = B$.



Question

If $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$ and $D = \{7, 8, 9, 10\}$; find

(i) $A \cup B$

(ii) $A \cup C$

(iii) $B \cup C$

(iv) $B \cup D$

(v) $A \cup B \cup C$

(vi) $A \cup B \cup D$

(vii) $B \cup C \cup D$

$$(v) A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$\Rightarrow A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

h/w

Question

Find the intersection of each pair of sets of question 1 above.

$$(1)(sol) \quad X = \{1, 3, 5\} \text{ \& } Y = \{1, 2, 3\}$$

$$\Rightarrow X \cap Y = \{1, 3\}$$

Question

If $A = \{3, 5, 7, 9, 11\}$, $B = \{7, 9, 11, 13\}$, $C = \{11, 13, 15\}$ and $D = \{15, 17\}$; find

(i) $A \cap B$

(ii) $B \cap C$

(iii) $A \cap C \cap D$

(iv) $A \cap C$

(v) $B \cap D$

(vi) $A \cap (B \cup C)$

(vii) $A \cap D$

(viii) $A \cap (B \cup D)$

(ix) $(A \cap B) \cap (B \cup C)$

(x) $(A \cap D) \cap (B \cup C)$

(vii) (Sol) $B \cup D = \{7, 9, 11, 13, 15, 17\}$

$\Rightarrow A \cap (B \cup D) = \{7, 9, 11\}$

H/w

Question



If

$A = \{x : x \text{ is a natural number}\},$

$B = \{x : x \text{ is an even natural number}\}$

$C = \{x : x \text{ is an odd natural number}\}$ and

$D = \{x : x \text{ is a prime number}\},$ find

✓ (i) $A \cap B$

H/W (ii) $A \cap C$

✓ (iii) $A \cap D$

H/W (iv) $B \cap C$

✓ (v) $B \cap D$

✓ (vi) $C \cap D$

(Sol) $A = \{1, 2, 3, 4, 5, 6, \dots\}$

$$B = \{2, 4, 6, 8, 10, \dots\}$$

$$C = \{1, 3, 5, 7, 9, 11, \dots\}$$

$$D = \{2, 3, 5, 7, 11, 13, \dots\}$$

So, (i) $A \cap B = \{2, 4, 6, 8, 10, \dots\} = B.$

(iii) $A \cap D = \{2, 3, 5, 7, 11, 13, \dots\}$
 $\Rightarrow A \cap D = D.$

Question

$$A \cap B = \phi$$

Which of the following pairs of sets are disjoint

(i) ~~X~~ $\{1, 2, 3, 4\}$ and $\{x : x \text{ is a natural number and } 4 \leq x \leq 6\} = \{4, 5, 6\}$

(ii) ~~X~~ $\{a, e, i, o, u\}$ and $\{c, d, e, f\}$

✓ (iii) $\{x : x \text{ is an even integer}\}$ and $\{x : x \text{ is an odd integer}\}$

$$\{\dots, -4, -2, 0, 2, 4, \dots\} \text{ \& \} \{\dots, -5, -3, -1, 1, 3, 5, \dots\}$$

Question

If $A = \{3, 6, 9, 12, 15, 18, 21\}$, $B = \{4, 8, 12, 16, 20\}$,
 $C = \{2, 4, 6, 8, 10, 12, 14, 16\}$, $D = \{5, 10, 15, 20\}$; find

(i) $A - B$

(ii) $A - C$

(iii) $A - D$

(iv) $B - A$

(v) $C - A$

(vi) $D - A$

(vii) $B - C$

(viii) $B - D$

(ix) $C - B$

(x) $D - B$

(xi) $C - D$

(xii) $D - C$

(Sol) (i) $A - B = \{3, 6, 9, 15, 18, 21\}$

(ii) $C - D = \{2, 4, 6, 8, 12, 14, 16\}$

Question

If $X = \{a, b, c, d\}$ and $Y = \{f, b, d, g\}$ find

(i) $X - Y = \{a, c\}$

(ii) $Y - X = \{f, g\}$

(iii) $X \cap Y = \{b, d\}$

Question

If $\mathbb{R}$ is the set of real numbers and $\mathbb{Q}$ is the set of rational numbers, then what is $\mathbb{R} - \mathbb{Q}$.

(sol) $\mathbb{R} - \mathbb{Q} = \text{Set of Irrational numbers.}$

$$A \setminus B = A - B$$

$$\mathbb{R} - \mathbb{Q} = \mathbb{R} \setminus \mathbb{Q}$$

Question

State whether each of the following statement is **true or false**. Justify your answer.

- (i) $\{2, 3, 4, 5\}$ and $\{3, 6\}$ are disjoint sets. (False)
- (ii) $\{a, e, \text{oo}, i, o, u\}$ and $\{a, b, c, d\}$ are disjoint sets. (False)
- (iii) $\{2, 6, 10, 14\}$ and $\{3, 7, 11, 15\}$ are disjoint sets. (True)
- (iv) $\{2, 6, 10\}$ and $\{3, 7, 11\}$ are disjoint sets. (True)

Complement of a Set

let A be any Set, then the Complement of A is denoted and defined as;

$$A' = U - A = \{x : x \in U \text{ but } x \notin A\}$$

For example: $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $A = \{2, 3, 4\}$, $B = \{6, 7, 8, 9\}$
 $\Rightarrow A' = \{1, 5, 6, 7, 8, 9, 10\}$

$$B' = \{1, 2, 3, 4, 5, 10\}$$



Properties of Complement Sets

- (i) **Complement Laws**: $\checkmark (a) A \cup A' = U$ $\checkmark (b) A \cap A' = \phi$
- (ii) **De Morgan's laws**: $\checkmark (b) (A \cup B)' = A' \cap B'$ $(b) (A \cap B)' = A' \cup B'$
- (iii) **Law of Double Decomposition**: $(A')' = A$
- (iv) **Law of empty set and universal set**: $\phi' = U$ and $U' = \phi$.

QUESTION (NCERT EXEMPLAR):

If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 5\}$, $B = \{2, 4, 6, 7\}$ and $C = \{2, 3, 4, 8\}$. Then

(i) $(B \cup C)'$ is _____.

(ii) $(C - A)'$ is _____.

$$(Sol)(i) B \cup C = \{2, 3, 4, 6, 7, 8\}$$

$$\Rightarrow (B \cup C)' = \{1, 5, 9, 10\}$$

$$(ii) C - A = \{4, 8\}$$

$$\Rightarrow (C - A)' = \{1, 2, 3, 5, 6, 7, 9, 10\}$$

QUESTION (NCERT EXEMPLAR):

$$U' = \phi$$



If X and Y are two sets and X' denotes the complement of X , then $X \cap (X \cup Y)'$ is equal to

☒ 1 X

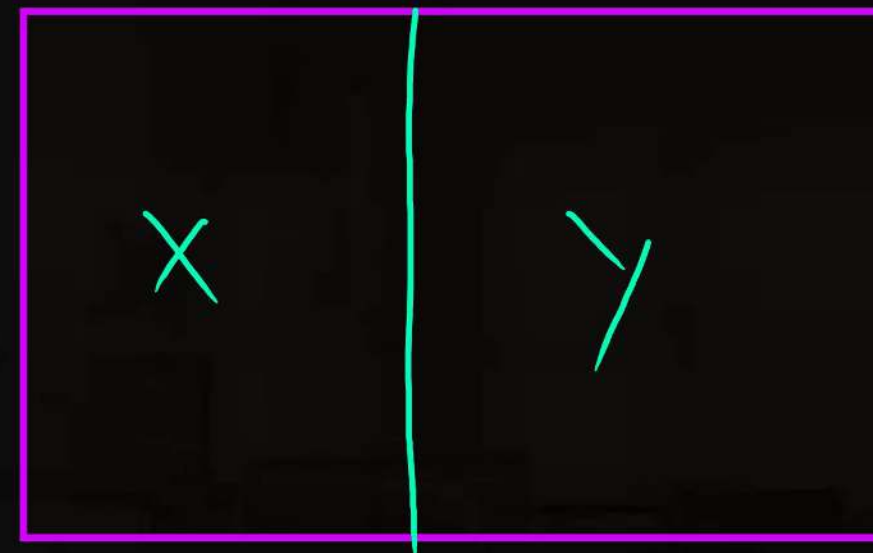
$X \cup Y \rightarrow$ universal set

☒ 2 Y

$$X \cap (X \cup Y)' = X \cap \phi = \phi$$

☒ 3 ϕ

☒ 4 $X \cap Y$



EXERCISE – 1.5

Question

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$. Find

(i) A'

(sol⁽ⁱ⁾) $A' = \{5, 6, 7, 8, 9\}$

(ii) B'

(ii^{oo}) $B' = \{1, 3, 5, 7, 9\}$

(iii) $(A \cup C)'$

(iii^{oo}) $A \cup C = \{1, 2, 3, 4, 5, 6\}$

(iv) $(A \cup B)'$

$\Rightarrow (A \cup C)' = \{7, 8, 9\}$

(v) $(A')'$

(v) $A' = \{5, 6, 7, 8, 9\}$

$\Rightarrow (A')' = \{1, 2, 3, 4\} = A$

(vi) $(B - C)'$

Question

If $U = \{a, b, c, d, e, f, g, h\}$, find the complements of the following sets:

✓ (i) $A = \{a, b, c\}$

✓ (ii) $B = \{d, e, f, g\}$

✓ (iii) $C = \{a, c, e, g\}$

✓ (iv) $D = \{f, g, h, a\}$

H/W.

Question

Taking the set of natural numbers as the universal set, write down the complements of the following sets:

- (i) $\{x : x \text{ is an even natural number}\} \longrightarrow \text{Set of odd natural numbers.}$
- (ii) $\{x : x \text{ is an odd natural number}\} \longrightarrow \text{Set of even natural numbers.}$
- (iii) $\{x : x \text{ is a positive multiple of 3}\} \longrightarrow \{x : x \text{ is not a positive multiple of 3}\}.$
- (iv) $\{x : x \text{ is a prime number}\}$
- (v) $\{x : x \text{ is a natural number divisible by 3 and 5}\} = \{15, 30, 45, \dots\}$
- (vi) $\{x : x \text{ is a perfect square}\}$
- (vii) $\{x : x \text{ is a perfect cube}\}$
- (viii) $\{x : x + 5 = 8\} \Rightarrow \{x : x + 5 \neq 8\} = \{x : x \neq 3\}$
- (ix) $\{x : 2x + 5 = 9\}$
- (x) $\{x : x \geq 7\} \rightarrow \{x : x < 7\} = \{1, 2, 3, 4, 5, 6\}.$
- (xi) $\{x : x \in \mathbb{N} \text{ and } 2x + 1 > 10\}$

Question

(PYQ JK BOSE)

If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$ and $B = \{2, 3, 5, 7\}$ Verify that

✓ (i) $(A \cup B)' = A' \cap B'$

✓ (ii) $(A \cap B)' = A' \cup B'$ (H/W)

(Sol) (i) $A' = \{1, 3, 5, 7, 9\}$

$B' = \{1, 4, 6, 8, 9\}$

$A \cup B = \{2, 3, 4, 5, 6, 7, 8\}$

So, $(A \cup B)' = \{1, 9\}$ — ①

& $A' \cap B' = \{1, 9\}$ — ②

From ① & ②, we get,

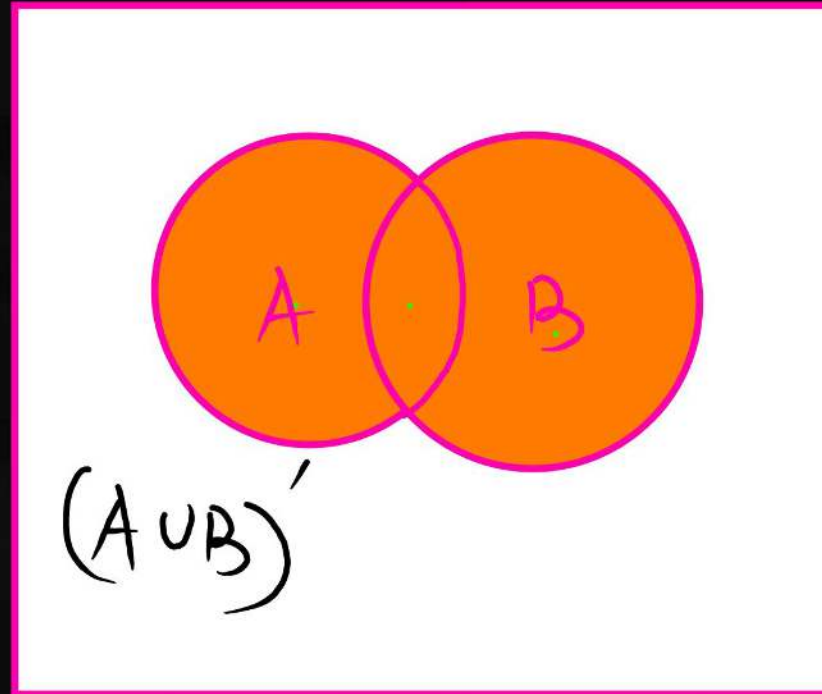
$(A \cup B)' = A' \cap B'$

Hence verified

Question

Draw appropriate Venn diagram for each of the following:

- ✓ (i) $(A \cup B)'$
- ✓ (ii) $A' \cap B'$
- ✓ (iii) $(A \cap B)'$
- ✓ (iv) $A' \cup B'$



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Question

Let U be the set of all triangles in a plane. If A is the set of all triangles with at least one angle different from 60° , what is A' ?

(Sol) $A' = \{x : x \text{ is an equilateral triangle}\}$

Question

Fill in the blanks to make each of the following a true statement:

(i) $A \cup A' = \dots$

(ii) $\phi' \cap A = \dots A = A$

(iii) $A \cap A' = \dots$

(iv) $U' \cap A = \dots A = \phi$

MISCELLANEOUS EXERCISE

Question

(JEE Mains)

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$



Let A , B , and C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$.

Show that $B = C$.

Proof: We know that

$$B = B \cap (A \cup B)$$

$$\Rightarrow B = B \cap (A \cup C)$$

By Distributive law;

$$B = (B \cap A) \cup (B \cap C)$$

$$\Rightarrow B = (A \cap B) \cup (B \cap C)$$

$$\Rightarrow B = (A \cap C) \cup (B \cap C)$$

$$\Rightarrow B = (C \cap A) \cup (C \cap B)$$

$$\Rightarrow B = C \cap (A \cup B) \text{ (Distributive law)}$$

$$\Rightarrow B = C \cap (A \cup C)$$

$$\Rightarrow B = C$$

Hence proved.

NCERT EXEMPLAR
Objective R.D.
Sharma.

Question

Show that $A \cap B = A \cap C$ need not imply $B = C$.

H/W

Question

Let A and B be sets. If $A \cap X = B \cap X = \phi$ and $A \cup X = B \cup X$ for some set X , show that $A = B$.

(**Hints:** $A = A \cap (A \cup X)$, $B = B \cap (B \cup X)$ and use Distributive law)

H/W

Question

Find sets A , B and C such that $A \cap B$, $B \cap C$ and $A \cap C$ are non-empty sets and $A \cap B \cap C = \phi$.

h/w

$$A = \{2, 4\}, B = \{2, 3\}, C = \{3\}$$

$$A \cup B = \{2, 3, 4\}$$

$$B \neq C.$$

$$A \cup C = \{2, 3, 4\}$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Chapter I ends

Thank
You